

Cash Generation Multiple and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Cash Generation Multiple (CGM), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on CGM achieves an annualized gross (net) Sharpe ratio of 0.53 (0.50), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 39 (39) bps/month with a t-statistic of 3.51 (3.55), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Cash-based operating profitability, Operating profitability RD adjusted, net income / book equity, Return on assets (qtrly), Cash-flow to price variance, Analyst earnings per share) is 19 bps/month with a t-statistic of 2.03.

1 Introduction

Asset prices reflect investors' marginal utility of consumption across states of nature, with expected returns compensating for systematic exposure to consumption risk. While the consumption-based asset pricing paradigm provides the theoretical foundation for understanding cross-sectional return variation, identifying firm characteristics that proxy for consumption risk exposure remains a central challenge in finance. We examine whether Cash Generation Multiple (CGM), defined as the ratio of operating cash flow to total accruals, captures firms' sensitivity to aggregate consumption fluctuations and thus predicts cross-sectional stock returns. The ability of firms to generate cash relative to their accrual-based earnings potentially signals their resilience during consumption downturns, when investors' marginal utility is high. Firms with low cash generation multiples may face greater distress during economic contractions, making their equity claims particularly sensitive to consumption risk.

The theoretical link between cash generation and consumption risk operates through firms' differential exposure to aggregate consumption shocks. Following the long-run risks framework of [Bansal and Yaron \(2004\)](#), persistent shocks to consumption growth generate substantial risk premia when investors have preferences for early resolution of uncertainty. Firms with low CGM ratios depend heavily on accrual accounting to report earnings, suggesting potential vulnerability when consumption growth slows and external financing becomes costly. These firms' cash flows covary more strongly with the stochastic discount factor during bad states when marginal utility is high, consistent with [Campbell and Cochrane \(1999\)](#) habit formation models where risk premia vary with the consumption surplus ratio. The state-dependent nature of CGM's predictive power aligns with disaster risk models [Barro \(2006\)](#), where firms with weak cash generation face elevated bankruptcy risk during consumption disasters. During normal times, differences in cash gener-

ation may matter less for equity valuations, but in bad states characterized by low consumption growth, investors demand higher returns from firms whose cash flows provide poor consumption hedges. The intertemporal marginal rate of substitution prices assets according to their covariance with consumption growth, and firms with low CGM exhibit stronger negative covariance during consumption downturns [Lettau and Ludvigson \(2001\)](#). This consumption-based mechanism predicts that sorting on CGM generates return spreads that compensate investors for bearing systematic consumption risk, particularly the risk associated with rare disasters and long-run consumption volatility.

The value-weighted long/short trading strategy based on CGM achieves an annualized gross Sharpe ratio of 0.53, with monthly average abnormal gross returns relative to the Fama-French six-factor model of 39 basis points per month (t-statistic = 3.51). The strategy’s performance remains robust across various portfolio construction methodologies, with twenty out of twenty-five alternative specifications generating alphas with t-statistics exceeding two. Among the largest stocks, those with market capitalization above the 80th NYSE percentile, the CGM strategy delivers average returns of 54 basis points per month (t-statistic = 3.09), demonstrating that the anomaly is not confined to small, illiquid securities where consumption risk effects might be confounded by market frictions. The strategy’s factor loadings reveal significant exposure to profitability factors, with a loading of 0.59 on RMW (t-statistic = 11.77), consistent with cash generation capturing firms’ operational efficiency during consumption downturns. After accounting for transaction costs using high-frequency bid-ask spreads, the net Sharpe ratio remains economically significant at 0.50, ranking in the 99th percentile among 212 documented anomalies. The CGM strategy maintains statistical significance even after controlling for six closely related anomalies, generating a monthly alpha of 19 basis points (t-statistic = 2.03) relative to the six Fama-French factors plus related profitability and cash flow measures.

Our findings contribute to the consumption-based asset pricing literature by identifying a firm characteristic that captures cross-sectional variation in consumption risk exposure. While [Parker and Julliard \(2005\)](#) demonstrate that ultimate consumption risk measured over longer horizons explains the cross-section of returns, and [Yogo \(2006\)](#) shows that durable consumption growth prices assets better than nondurable consumption, we provide evidence that CGM serves as an observable proxy for firms’ consumption risk exposure. Unlike [Savov \(2011\)](#), who uses garbage as a measure of consumption to address measurement error, our approach identifies a firm-level characteristic that predicts which stocks provide poor consumption hedges. The CGM signal complements the profitability anomaly documented in [Novy-Marx \(2013\)](#) by emphasizing the cash component of earnings that matters most during consumption disasters. Our results extend [Hou et al. \(2015\)](#)’s investment-based explanation for profitability by linking cash generation directly to firms’ ability to maintain operations when the marginal utility of consumption is high. Methodologically, we employ the comprehensive testing framework of [Novy-Marx and Velikov \(2023\)](#) to establish that CGM’s predictive power survives extensive robustness tests and transaction costs, addressing concerns raised by [Hou et al. \(2020\)](#) about data mining in the anomalies literature. The economic magnitude of the CGM effect, particularly among large-cap stocks where consumption-based theories should dominate behavioral explanations, supports the rational risk interpretation of cross-sectional return predictability.

2 Data

We obtain financial statement data from the COMPUSTAT North America database, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all firms with available data for the required variables,

spanning several decades of financial reporting. We focus on annual observations to ensure consistency in the measurement of cash flow and depreciation metrics across firms. Cash Generation Multiple combines two key financial statement items. Operating activities net cash flow (OANCF) represents the cash generated from a company’s core business operations, calculated as net income adjusted for non-cash charges and changes in working capital. This measure captures the actual cash a firm produces through its primary business activities during the fiscal year. Depreciation, depletion, and amortization (DPACT) reflects the non-cash expense that allocates the cost of tangible and intangible assets over their useful lives, representing the accounting recognition of asset consumption in the production process. We construct the Cash Generation Multiple by dividing operating activities net cash flow (OANCF) by depreciation, depletion, and amortization (DPACT). This ratio captures the relationship between cash generated from operations and the non-cash charges associated with asset utilization, providing insight into how efficiently a firm converts its productive asset base into operating cash flows. We calculate the signal using fiscal year-end values to maintain temporal consistency across firms with different fiscal year endings. To ensure data quality, we require non-missing values for both OANCF and DPACT, and exclude observations where DPACT equals zero to avoid undefined ratios. The resulting measure provides a standardized metric that facilitates cross-sectional comparisons of cash generation efficiency relative to the depreciation of productive assets.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the CGM signal. Panel A plots the time-series of the mean, median, and interquartile range for CGM. On average, the cross-sectional mean (median) CGM is -5.38 (0.29) over the 1988 to 2024 sample, where the

starting date is determined by the availability of the input CGM data. The signal's interquartile range spans -3.16 to 1.00. Panel B of Figure 1 plots the time-series of the coverage of the CGM signal for the CRSP universe. On average, the CGM signal is available for 6.67% of CRSP names, which on average make up 7.44% of total market capitalization.

4 Does CGM predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on CGM using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high CGM portfolio and sells the low CGM portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short CGM strategy earns an average return of 0.50% per month with a t-statistic of 3.17. The annualized Sharpe ratio of the strategy is 0.53. The alphas range from 0.39% to 0.64% per month and have t-statistics exceeding 3.51 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is 0.59, with a t-statistic of 11.77 on the RMW factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 489 stocks and an average market capitalization of at least \$1,080 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals 37 bps/month with a t-statistics of 2.03. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty exceed two, and for seventeen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between 20-55bps/month. The lowest return, (20 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of 1.10. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the CGM trading

strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-three cases, and significantly expands the achievable frontier in seventeen cases.

Table 3 provides direct tests for the role size plays in the CGM strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and CGM, as well as average returns and alphas for long/short trading CGM strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the CGM strategy achieves an average return of 54 bps/month with a t-statistic of 3.09. Among these large cap stocks, the alphas for the CGM strategy relative to the five most common factor models range from 34 to 59 bps/month with t-statistics between 2.01 and 4.14.

5 How does CGM perform relative to the zoo?

Figure 2 puts the performance of CGM in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the CGM strategy falls in the distribution. The CGM strategy’s gross (net) Sharpe ratio of 0.53 (0.50) is greater than 92% (99%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the CGM strategy (red line).² Ignoring trading costs, a \$1 invested in the CGM strategy

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides

would have yielded \$5.92 which ranks the CGM strategy in the top 0% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the CGM strategy would have yielded \$5.19 which ranks the CGM strategy in the top 0% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the CGM relative to those. Panel A shows that the CGM strategy gross alphas fall between the 83 and 94 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 198806 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The CGM strategy has a positive net generalized alpha for five out of the five factor models. In these cases CGM ranks between the 96 and 98 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does CGM add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations.

Figure 5 plots a name histogram of the correlations of CGM with 210 filtered anomaly of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price CGM or at least to weaken the power CGM has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of CGM conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{CGM} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{CGM}CGM_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{CGM,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on CGM. Stocks are finally grouped into five CGM portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted CGM trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on CGM and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the CGM signal in these Fama-MacBeth regressions exceed -0.70,

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

with the minimum t-statistic occurring when controlling for Cash-based operating profitability. Controlling for all six closely related anomalies, the t-statistic on CGM is 0.22.

Similarly, Table 5 reports results from spanning tests that regress returns to the CGM strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the CGM strategy earns alphas that range from 21-37bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.07, which is achieved when controlling for Cash-based operating profitability. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the CGM trading strategy achieves an alpha of 19bps/month with a t-statistic of 2.03.

7 Does CGM add relative to the whole zoo?

Finally, we can ask how much adding CGM to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 158 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 158 anomalies augmented with the CGM signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 158-anomaly com-

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which CGM is available.

bination strategy grows to \$36.86, while \$1 investment in the combination strategy that includes CGM grows to \$41.84.

8 Conclusion

This study provides compelling evidence that the Cash Generation Multiple (CGM) represents a robust and economically significant predictor of cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that CGM-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for well-established risk factors and related anomalies.

The key findings reveal that a value-weighted long/short strategy based on CGM delivers impressive performance metrics, with an annualized Sharpe ratio of 0.53 gross and 0.50 net of transaction costs. The strategy’s resilience is particularly noteworthy, maintaining statistical and economic significance with monthly alphas of 39 basis points relative to the Fama-French five-factor model plus momentum (t-statistic = 3.55). Even more striking is the signal’s ability to generate positive abnormal returns when evaluated against a comprehensive set of twelve factors, including six closely related profitability and cash flow measures, yielding a monthly alpha of 19 basis points (t-statistic = 2.03).

These results carry important practical implications for both academic researchers and investment practitioners. The persistence of CGM’s predictive power, even after accounting for transaction costs and controlling for related factors, suggests that cash generation metrics capture unique information about firm fundamentals not fully reflected in existing factor models. For practitioners, the signal’s robust performance and relatively modest degradation from gross to net returns indicate potential implementability in real-world portfolio construction.

Despite these encouraging findings, several limitations warrant consideration. First, while our analysis follows state-of-the-art protocols for signal evaluation, the true out-of-sample performance of CGM-based strategies remains uncertain given the proliferation of factors in the academic literature. Second, the study focuses primarily on U.S. equities, and the signal’s efficacy in international markets requires further investigation. Third, the economic mechanisms underlying CGM’s predictive power deserve deeper theoretical and empirical exploration.

Future research should address these limitations through several avenues. International replication studies would establish the global applicability of CGM as a return predictor. Time-series analysis of the signal’s performance across different market regimes and economic cycles could provide insights into its conditional effectiveness. Additionally, investigating the interaction between CGM and corporate events such as earnings announcements, dividend changes, or capital structure decisions may illuminate the economic channels through which cash generation metrics influence asset prices. Finally, machine learning techniques could be employed to explore potential non-linearities in the CGM-return relationship and optimize signal construction for enhanced predictive power.

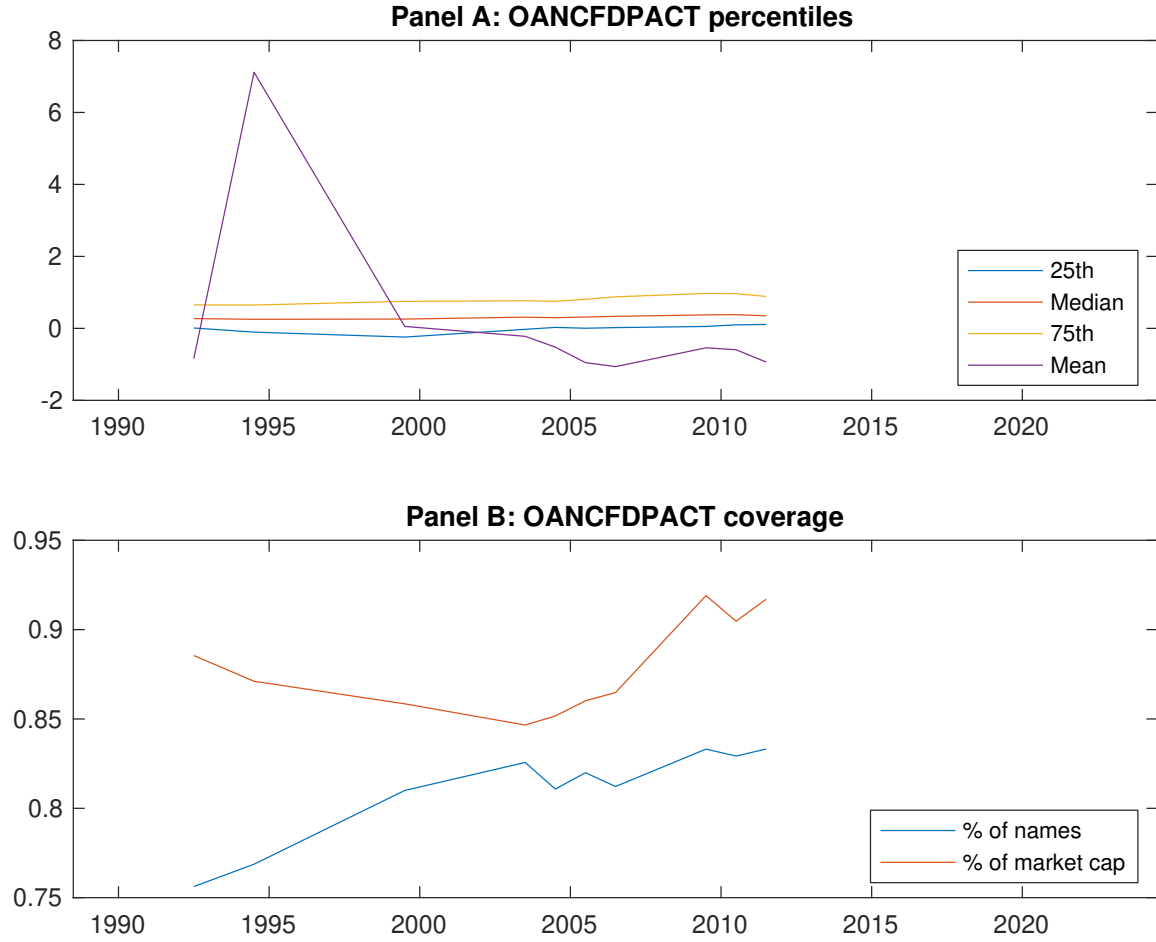


Figure 1: Times series of CGM percentiles and coverage. This figure plots descriptive statistics for CGM. Panel A shows cross-sectional percentiles of CGM over the sample. Panel B plots the monthly coverage of CGM relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on CGM. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, [Fama and French \(1993\)](#) three-factor model, [Fama and French \(1993\)](#) three-factor model augmented with the [Carhart \(1997\)](#) momentum factor, [Fama and French \(2015\)](#) five-factor model, and the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor following [Fama and French \(2018\)](#). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the [Fama and French \(2015\)](#) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 198806 to 202406.

Panel A: Excess returns and alphas on CGM-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.45 [1.53]	0.60 [3.13]	0.74 [3.52]	0.70 [3.50]	0.95 [4.03]	0.50 [3.17]
α_{CAPM}	-0.49 [-3.80]	0.01 [0.14]	0.04 [0.54]	0.02 [0.34]	0.15 [2.22]	0.64 [4.06]
α_{FF3}	-0.43 [-4.43]	-0.05 [-0.55]	0.02 [0.25]	0.01 [0.26]	0.20 [3.78]	0.63 [5.19]
α_{FF4}	-0.39 [-3.99]	-0.05 [-0.62]	0.03 [0.46]	0.02 [0.36]	0.21 [3.90]	0.60 [4.88]
α_{FF5}	-0.18 [-2.19]	-0.21 [-2.51]	-0.09 [-1.26]	-0.07 [-1.28]	0.23 [4.34]	0.41 [3.75]
α_{FF6}	-0.16 [-1.88]	-0.20 [-2.40]	-0.07 [-0.95]	-0.06 [-1.12]	0.23 [4.34]	0.39 [3.51]
Panel B: Fama and French (2018) 6-factor model loadings for CGM-sorted portfolios						
β_{MKT}	1.10 [52.03]	0.88 [41.67]	0.99 [56.96]	0.96 [73.04]	1.05 [79.99]	-0.04 [-1.54]
β_{SMB}	0.37 [12.24]	0.04 [1.33]	0.08 [3.05]	-0.04 [-2.24]	-0.08 [-4.49]	-0.45 [-11.51]
β_{HML}	0.01 [0.37]	0.11 [2.91]	-0.02 [-0.72]	-0.07 [-3.00]	-0.20 [-8.79]	-0.22 [-4.49]
β_{RMW}	-0.56 [-14.72]	0.19 [4.88]	0.12 [3.78]	0.17 [6.99]	0.03 [1.05]	0.59 [11.77]
β_{CMA}	0.01 [0.16]	0.36 [6.76]	0.25 [5.62]	0.04 [1.20]	-0.17 [-5.19]	-0.18 [-2.60]
β_{UMD}	-0.04 [-2.11]	-0.01 [-0.64]	-0.03 [-2.14]	-0.01 [-1.02]	-0.00 [-0.34]	0.04 [1.46]
Panel C: Average number of firms (n) and market capitalization (me)						
n	1373	489	516	582	770	
me (\$10 ⁶)	1080	1938	2560	3905	5768	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the CGM strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 198806 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.50 [3.17]	0.64 [4.06]	0.63 [5.19]	0.60 [4.88]	0.41 [3.75]	0.39 [3.51]
Quintile	NYSE	EW	0.37 [2.03]	0.51 [2.83]	0.43 [2.76]	0.31 [1.98]	0.15 [1.00]	0.05 [0.32]
Quintile	Name	VW	0.50 [2.03]	0.85 [3.67]	0.71 [4.21]	0.68 [3.95]	0.26 [1.76]	0.25 [1.67]
Quintile	Cap	VW	0.45 [3.03]	0.32 [2.19]	0.39 [3.24]	0.42 [3.40]	0.53 [4.63]	0.53 [4.60]
Decile	NYSE	VW	0.59 [3.05]	0.78 [4.16]	0.72 [4.82]	0.70 [4.63]	0.51 [3.69]	0.49 [3.54]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.48 [3.01]	0.60 [3.85]	0.57 [4.76]	0.56 [4.60]	0.40 [3.71]	0.39 [3.55]
Quintile	NYSE	EW	0.20 [1.10]	0.33 [1.78]	0.26 [1.59]	0.19 [1.17]		
Quintile	Name	VW	0.47 [1.88]	0.82 [3.53]	0.67 [3.94]	0.65 [3.80]	0.27 [1.84]	0.26 [1.77]
Quintile	Cap	VW	0.43 [2.91]	0.28 [1.95]	0.34 [2.83]	0.36 [2.97]	0.47 [4.15]	0.48 [4.16]
Decile	NYSE	VW	0.55 [2.86]	0.75 [3.98]	0.67 [4.49]	0.66 [4.40]	0.48 [3.49]	0.47 [3.41]

Table 3: Conditional sort on size and CGM

This table presents results for conditional double sorts on size and CGM. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on CGM. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high CGM and short stocks with low CGM. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 198806 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	CGM Quintiles					CGM Strategies						
	(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}	
	(1)	0.05 [0.11]	0.56 [1.41]	0.89 [2.84]	1.01 [3.36]	0.93 [3.17]	0.88 [3.35]	1.08 [4.19]	0.90 [4.20]	0.85 [3.92]	0.47 [2.44]	0.43 [2.24]
	(2)	0.32 [0.78]	0.64 [2.16]	0.92 [3.39]	0.87 [3.18]	0.91 [3.31]	0.59 [2.69]	0.89 [4.31]	0.74 [4.39]	0.67 [3.90]	0.31 [2.11]	0.26 [1.75]
	(3)	0.38 [1.10]	0.79 [3.13]	0.82 [3.17]	0.92 [3.53]	0.94 [3.36]	0.56 [3.41]	0.74 [4.61]	0.69 [4.44]	0.59 [3.80]	0.41 [2.82]	0.34 [2.32]
	(4)	0.56 [1.92]	0.89 [3.92]	0.75 [3.03]	0.86 [3.65]	0.98 [3.63]	0.43 [3.03]	0.47 [3.28]	0.49 [3.43]	0.42 [2.90]	0.40 [2.77]	0.33 [2.33]
	(5)	0.44 [2.10]	0.70 [3.61]	0.66 [3.41]	0.92 [4.36]	0.98 [3.67]	0.54 [3.09]	0.34 [2.01]	0.44 [3.00]	0.46 [3.12]	0.59 [4.14]	0.59 [4.10]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	CGM Quintiles					CGM Quintiles						
	Average n					Average market capitalization (\$10 ⁶)						
	(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)		
	(1)	399	401	408	413	412	37	34	43	54	58	
	(2)	123	123	124	123	123	79	83	86	86	87	
	(3)	83	84	84	83	83	139	145	147	148	147	
	(4)	70	70	70	70	70	305	320	317	326	329	
(5)	62	62	62	62	62	1774	2119	2410	3265	2713		

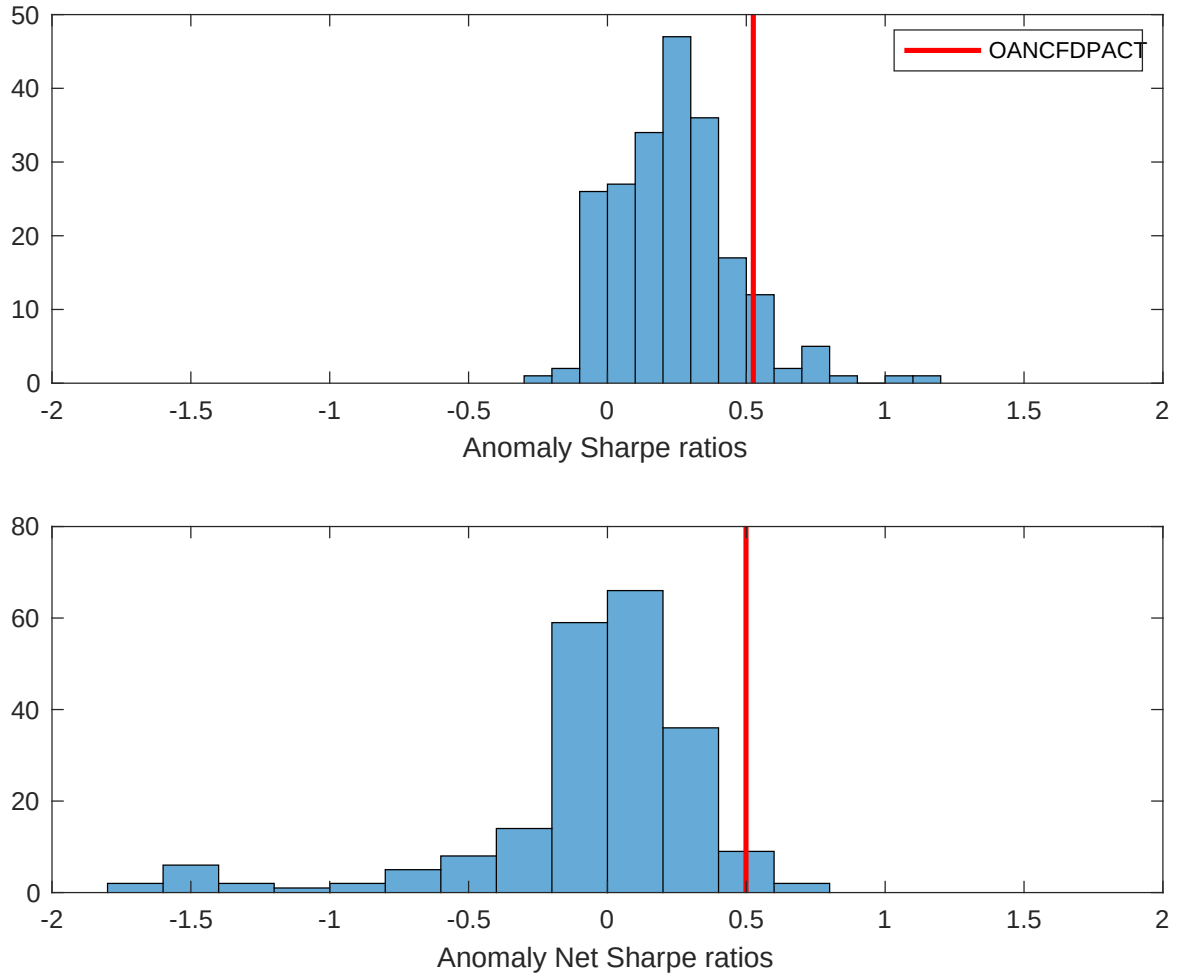


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the CGM with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

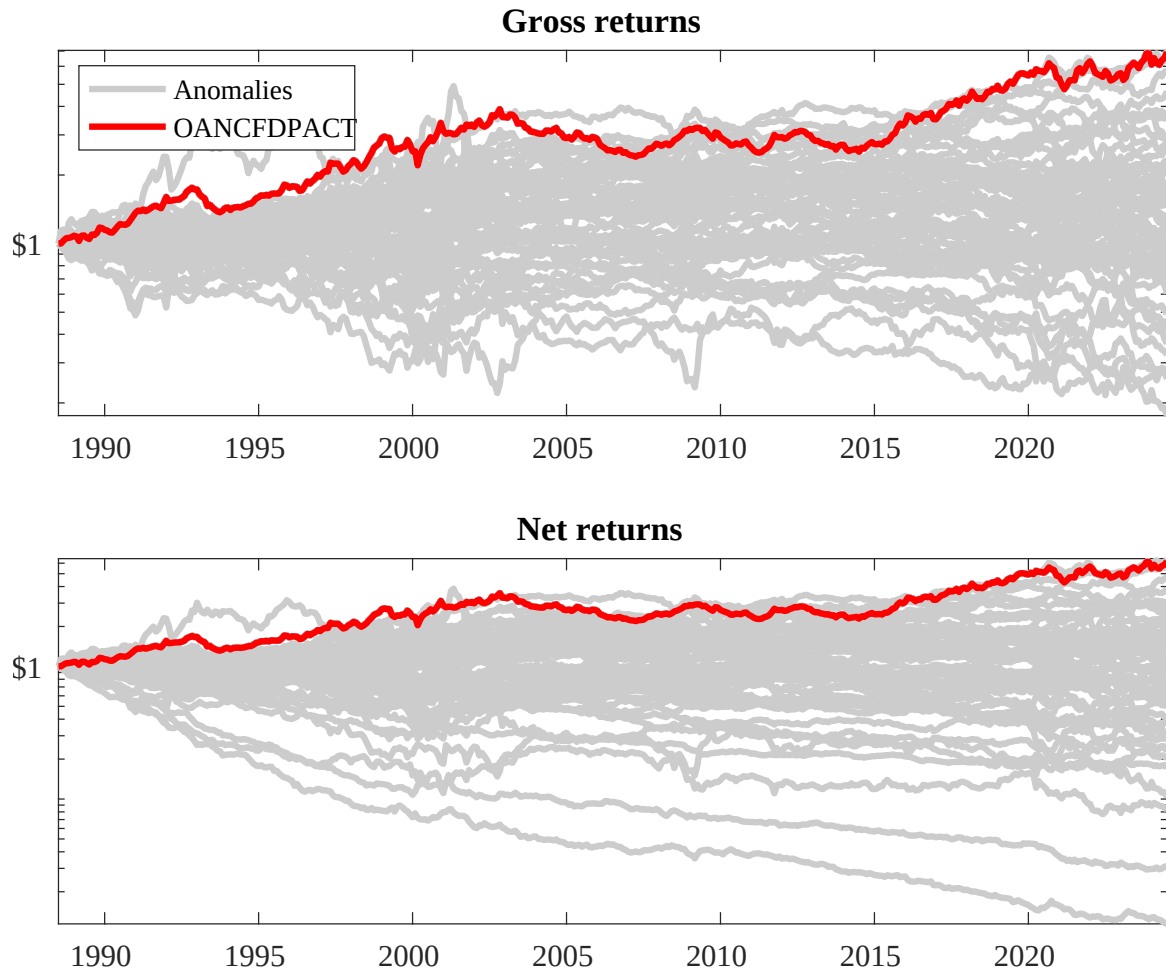
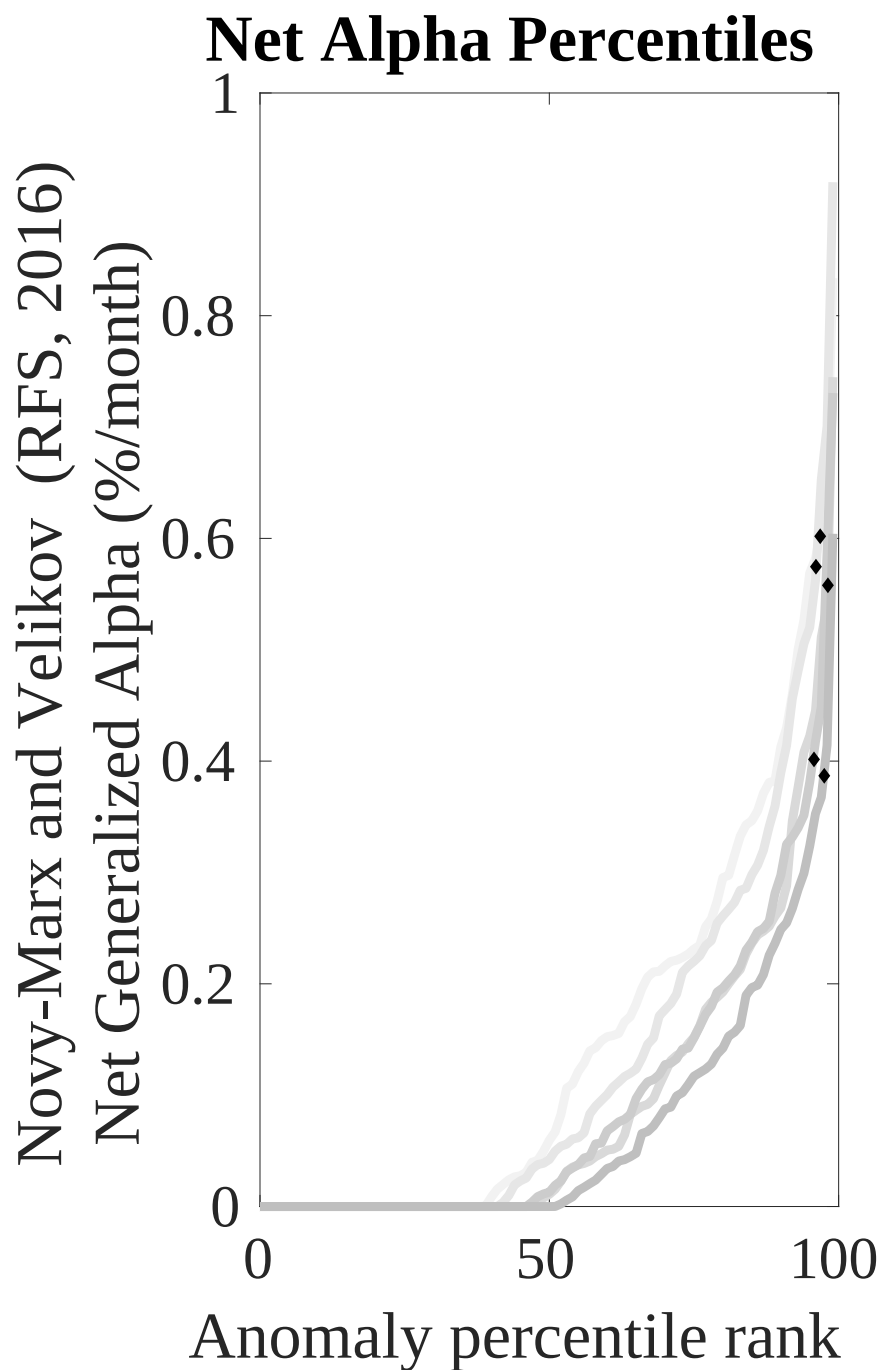
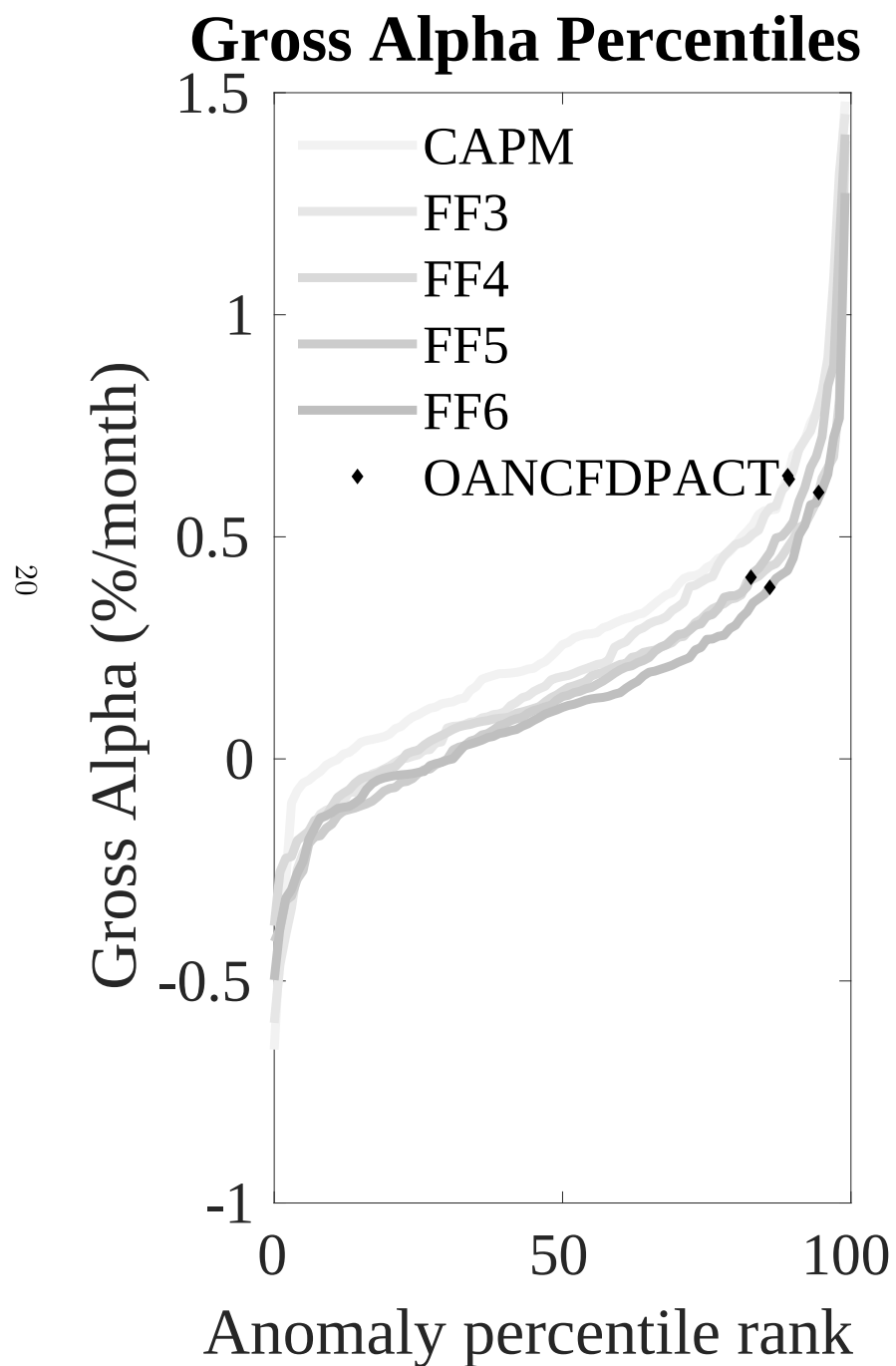


Figure 3: Dollar invested.

This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the CGM trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.



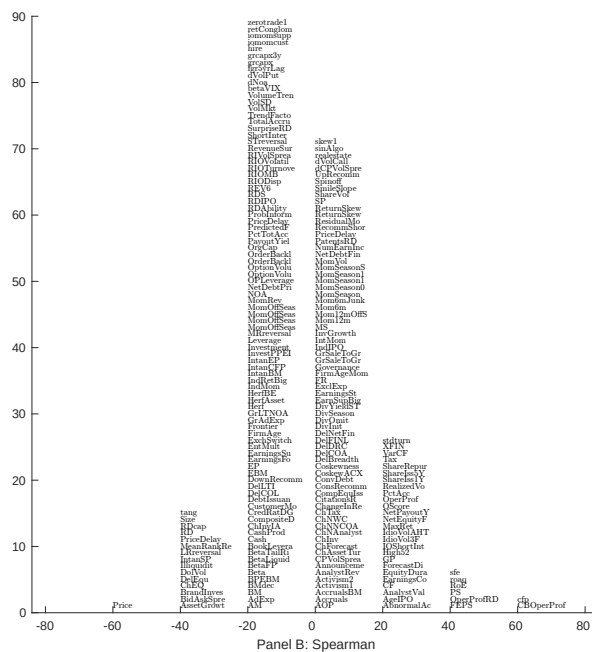
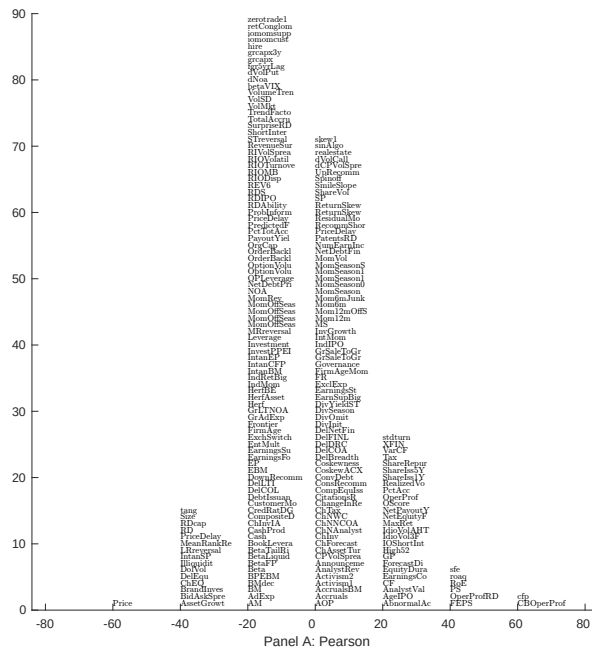


Figure 5: Distribution of correlations.

This figure plots a name histogram of correlations of 210 filtered anomaly signals with CGM. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

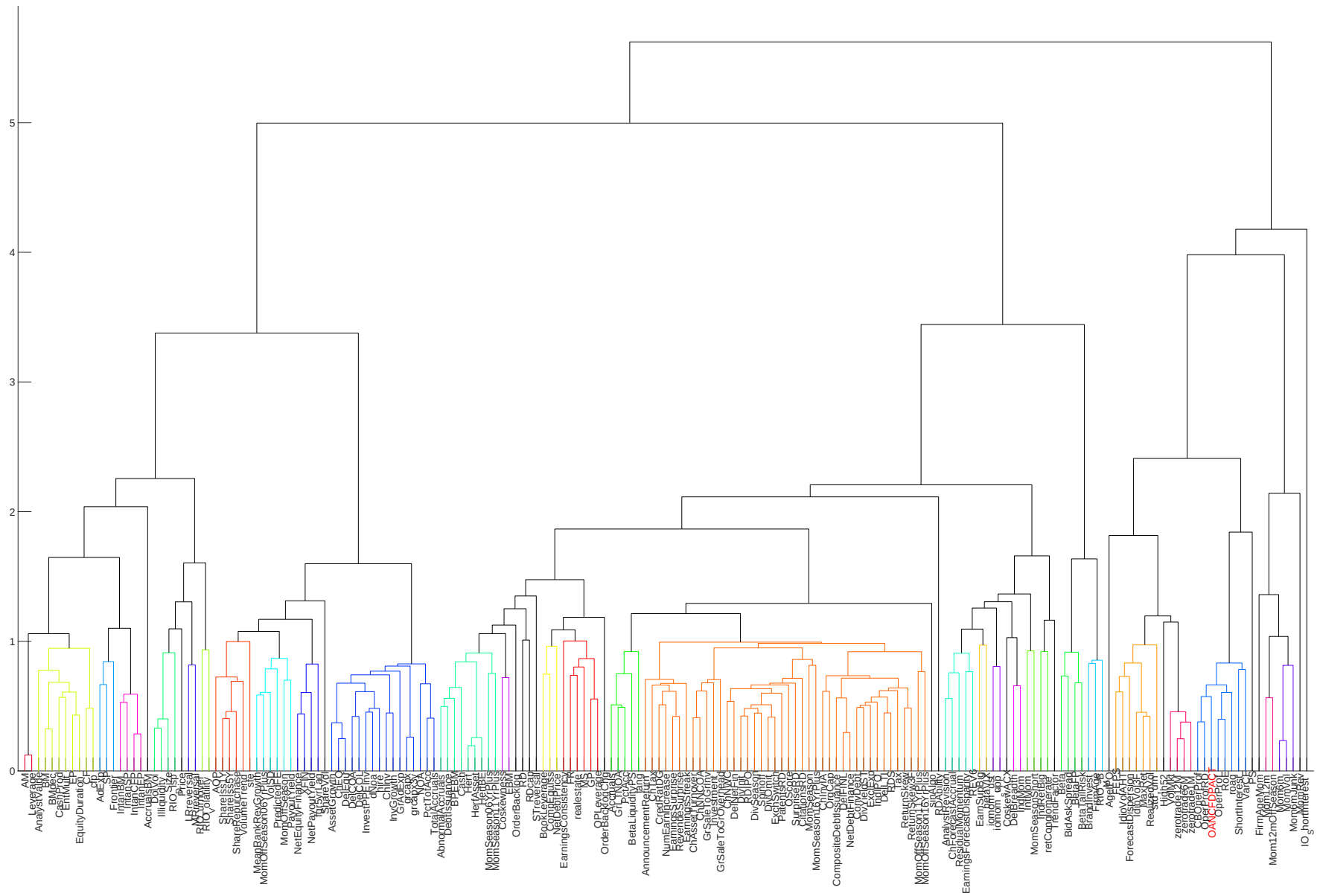


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

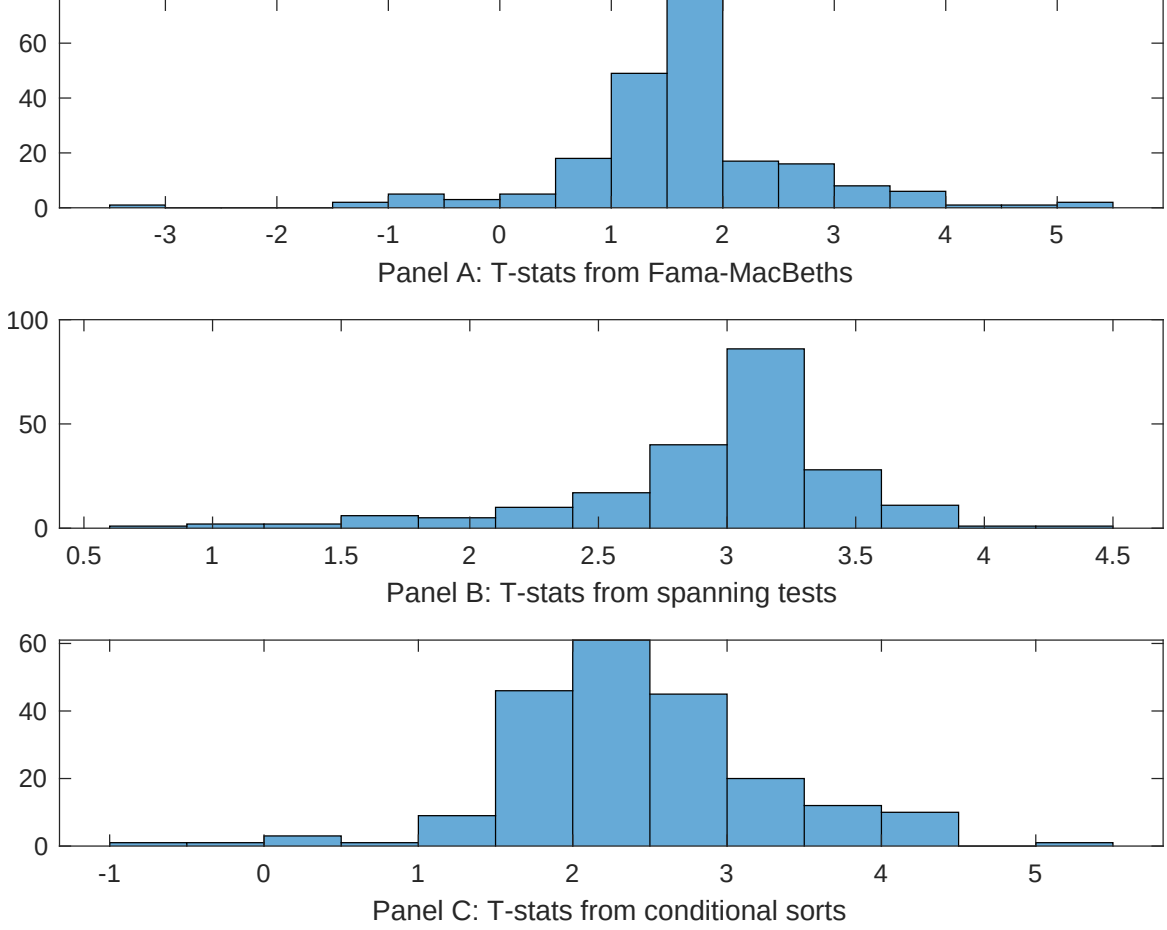


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of CGM conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{CGM} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{CGM}CGM_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{CGM,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on CGM. Stocks are finally grouped into five CGM portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted CGM trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on CGM. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{CGM}CGM_{i,t} + \sum_{k=1}^s \beta_{X_k} X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Cash-based operating profitability, Operating profitability RD adjusted, net income / book equity, Return on assets (qtrly), Cash-flow to price variance, Analyst earnings per share. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 198806 to 202406.

Intercept	0.94 [2.98]	0.91 [2.74]	0.11 [3.72]	0.12 [4.09]	0.12 [4.15]	0.93 [2.70]	0.85 [2.66]
CGM	-0.63 [-0.70]	0.87 [1.26]	0.17 [1.85]	0.54 [0.96]	0.18 [1.67]	0.26 [3.43]	0.24 [0.22]
Anomaly 1	0.20 [5.03]						0.15 [2.97]
Anomaly 2		0.17 [3.58]					0.30 [0.50]
Anomaly 3			0.56 [0.95]				-0.40 [-0.48]
Anomaly 4				0.44 [2.82]			0.24 [1.68]
Anomaly 5					0.12 [0.54]		0.31 [1.27]
Anomaly 6						0.40 [0.58]	-0.28 [-0.53]
# months	427	427	432	427	427	427	427
$\bar{R}^2(\%)$	1	1	1	1	1	2	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the CGM trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{CGM} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Cash-based operating profitability, Operating profitability RD adjusted, net income / book equity, Return on assets (qtrly), Cash-flow to price variance, Analyst earnings per share. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 198806 to 202406.

Intercept	0.21 [2.07]	0.22 [2.07]	0.36 [3.36]	0.33 [3.03]	0.34 [3.32]	0.37 [3.32]	0.19 [2.03]
Anomaly 1	45.59 [10.48]						32.75 [5.31]
Anomaly 2		35.25 [8.30]					5.46 [0.87]
Anomaly 3			35.50 [5.48]				8.87 [1.35]
Anomaly 4				23.91 [5.01]			11.28 [2.41]
Anomaly 5					31.12 [7.79]		21.51 [5.59]
Anomaly 6						2.66 [0.63]	-10.73 [-2.67]
mkt	0.02 [0.01]	1.87 [0.71]	1.12 [0.40]	-0.34 [-0.12]	4.90 [1.75]	-3.13 [-1.08]	6.25 [2.36]
smb	-32.62 [-8.88]	-34.21 [-8.92]	-32.61 [-7.51]	-38.84 [-9.81]	-23.46 [-5.20]	-42.01 [-8.93]	-21.72 [-4.73]
hml	-2.73 [-0.59]	-6.09 [-1.27]	-17.13 [-3.64]	-12.68 [-2.57]	-6.67 [-1.38]	-20.94 [-4.31]	10.36 [2.13]
rmw	35.95 [7.26]	32.86 [5.88]	27.80 [3.75]	37.22 [5.72]	44.23 [8.79]	55.99 [8.22]	22.10 [3.06]
cma	-27.66 [-4.39]	-20.82 [-3.20]	-14.99 [-2.21]	-20.26 [-2.97]	-9.05 [-1.36]	-19.69 [-2.76]	-15.09 [-2.40]
umd	-0.08 [-0.04]	-0.88 [-0.38]	1.73 [0.73]	-2.38 [-0.89]	-2.22 [-0.92]	3.17 [1.22]	-4.89 [-2.02]
# months	428	428	432	428	428	428	428
$\bar{R}^2(\%)$	66	63	60	59	62	57	69

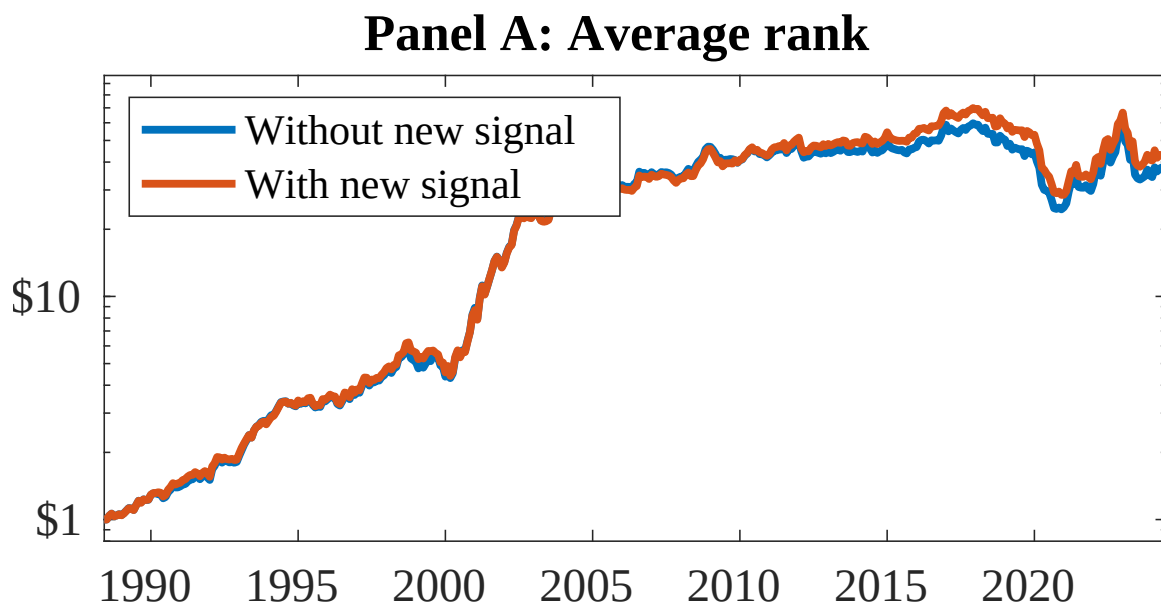


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 158 anomalies. The red solid lines indicate combination trading strategies that utilize the 158 anomalies as well as CGM. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

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